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FIDUCIAL DESIGNS AND RELATED METHODS FOR WIRE BOND PATTERN RECOGNITION

Abstract

Implementations of a semiconductor device may include a first die pad having a first metal layer thereon; and a fiducial for aligning to the semiconductor device, the fiducial including a portion including a photodefinable material formed over the first metal layer of the first die pad.

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Background/Summary

BACKGROUND

1. Technical Field

[0001] Aspects of this document relate generally to semiconductor devices and methods of manufacturing thereof that improve wire bond pattern recognition. More specific implementations involve including a fiducial on a die pad.

2. Background

[0002] Semiconductor devices include integrated circuits found in common electrical and electronic devices, such as phones, desktops, tablets, other computing devices, and other electronic devices. Semiconductor packages have been devised that can provide mechanical support and electrical connections to semiconductor devices included therein. Some of these semiconductor packages also provide electrostatic discharge protection and/or humidity protection for the devices.

SUMMARY

[0003] Implementations of a semiconductor device may include a first die pad having a first metal layer thereon; and a fiducial for aligning to the semiconductor device, the fiducial including a portion including a photodefinable material formed over the first metal layer of the first die pad.

[0004] Implementations of a semiconductor device may include one, all, or any of the following:

[0005] The semiconductor device may include a second die pad adjacent the first die pad, the second die pad having a second metal layer deposited thereon; and a second fiducial formed over the second metal layer. [0006] The semiconductor device may include a space between the first die pad and the second die pad where the second fiducial includes the space. [0007] The fiducial may be formed over an active portion of the first die pad. [0008] The semiconductor device may include a second fiducial for aligning to the semiconductor device, the second fiducial including a photodefinable material formed over the first metal layer of the first die pad. [0009] The fiducial may be triangular. [0010] The fiducial may include a metal alignment feature. [0011] The metal alignment feature may be part of the first metal layer. [0012] The metal alignment feature may be square. [0013] The metal alignment feature may be stepped. [0014] The metal alignment feature may be rectangular. [0015] The metal alignment feature may be triangular. [0016] The metal alignment feature may be spaced apart from an edge of the fiducial. [0017] The photodefinable material may border at least one edge of the fiducial. [0018] The photodefinable material may border at least two edges of the fiducial.

[0019] Implementations of a semiconductor device may include a first die pad including a first metal layer; and a second die pad adjacent the first die pad, the second die pad including a second metal layer; and a first fiducial for aligning the semiconductor device, the first fiducial including a photodefinable material formed over one of the first metal layer, the second metal layer, or both the first metal layer and the second metal layer.

[0020] Implementation of a semiconductor device may include one, all, or any of the following:

[0021] The first fiducial may include a portion of the first metal layer and a portion of the second metal layer. [0022] The semiconductor device may include a second fiducial for aligning the semiconductor device, the second fiducial including a photodefinable material formed over the first metal layer, the second metal layer, or both the first metal layer and the second metal layer. [0023] The semiconductor device may include a third fiducial for aligning the semiconductor device, the third fiducial including a photodefinable material formed over the first metal layer, the second metal layer, or both the first metal layer and the second metal layer.

[0024] Implementations of a semiconductor device may include a first die pad; and a second die pad adjacent the first die pad; and a first fiducial for aligning the semiconductor device, the first fiducial including a material formed over one of the first die pad, the second die pad, or both the first die pad and the second die pad.

[0025] The foregoing and other aspects, features, and advantages will be apparent to those artisans of ordinary skill in the art from the DESCRIPTION and DRAWINGS, and from the CLAIMS.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0026] Implementations will hereinafter be described in conjunction with the appended drawings, where like designations denote like elements, and:

[0027] FIG. 1 is a top view of an implementation of a semiconductor device including a plurality of fiducials;

[0028] FIG. 2 is an enlarged view of plurality of fiducials of FIG. 1;

[0029] FIG. 3 is a top view of an implementation of a fiducial on a semiconductor device;

[0030] FIG. 4 is a cross sectional side view of the semiconductor device of FIG. 3;

[0031] FIG. 5 is a top view of an implementation of fiducials on a semiconductor device;

[0032] FIG. 6 is a top view of an implementation of fiducials on a semiconductor device;

[0033] FIG. 7 is a top view of an implementation of fiducials on a semiconductor device;

[0034] FIG. 8 is a top view of an implementation of fiducials on a semiconductor device;

[0035] FIG. 9 is an enlarged partial top view of an implementation of fiducials on a semiconductor device; and

[0036] FIG. 10 is a top view of the implementation of fiducials on the semiconductor device of FIG. 9.

DESCRIPTION

[0037] This disclosure, its aspects and implementations, are not limited to the specific components, assembly procedures or method elements disclosed herein. Many additional components, assembly procedures and/or method elements known in the art consistent with the intended semiconductor devices and systems and methods thereof will become apparent for use with particular implementations from this disclosure. Accordingly, for example, although particular implementations are disclosed, such implementations and implementing components may comprise any shape, size, style, type, model, version, measurement, concentration, material, quantity, method element, step, and/or the like as is known in the art for such semiconductor devices and systems and methods thereof, and implementing components and methods, consistent with the intended operation and methods.

[0038] During fabrication, wire bonding is the process of bonding wires to the semiconductor devices to connect the semiconductor devices to packaging substrates. Wire bonding machines must be able to accurately place each wire on the semiconductor devices and packaging substrates. Pattern recognition is implemented to identify the proper locations for the wire bond on each semiconductor device to ensure alignment to each semiconductor device takes place properly prior to the wire bonding process.

[0039] While wirebonding generally forms electrical connections, in the semiconductor device examples provided herein, the wire bonds are connected on one end to the source pads, also known as die pads, of the semiconductor device and to a substrate or other semiconductor package component on the other end. Other wirebonds are also used to connect other device structures like the gate interconnect to the substrate or other semiconductor package. While the semiconductor device implementations disclosed herein are mostly single devices, such as, by non-limiting example, power transistors, power diodes, insulated gate bipolar transistors (IGBTs), silicon controlled rectifiers (SCRs), or other power semiconductor devices, the principles disclosed herein can also be used to form fiducials for a wide variety of other semiconductor device types, such as, by non-limiting example, microprocessors, controllers, microcontrollers, random access memories, field programmable gate arrays, flash memories, or any other semiconductor device type where wirebonding is employed as an interconnect.

[0040] When the wire bonds are not properly connected to the semiconductor source pads, particularly where they bridge or cause shorts to other pads, the semiconductor packages will not function properly. In practice, the wire bonds may be improperly connected to the substrate when the wire bonds are outside the desired source pad area, improperly connected to bus lines, or

improperly connected to two source pads, all of which cause electrical shorts or failures.

[0041] In wirebonding machines, pattern recognition techniques, software and equipment are used to detect a pattern that may include, for example, a fiducial or a plurality of fiducials on a semiconductor device, to accommodate for variations in alignment of each semiconductor device so a wire bond may be placed in a desired location on the source pads.

[0042] Fiducials are embedded objects, surface marks, or optically transmissive/partially optically transmissive layers that serve as a known reference point on a semiconductor device. A set of fiducials is used during alignment processing for image and pattern recognition techniques and systems used in wirebonders. The pattern recognition systems use the placement of the set of fiducials as reference points to determine the alignment and/or adjust the alignment of the semiconductor die itself so that the pre-programmed coordinates the wirebonder uses to place the wirebonds lead to formation of wirebonds at the desired locations on the semiconductor device. The fiducials aid in ensuring the semiconductor device or the wirebonding head(s) is oriented properly and so variations in direction and skew are comprehended.

[0043] With regard to the wire bonding process, slight variations in the orientation of the semiconductor device with respect to the wire bonding equipment may be sufficient to ruin the electrical connections of the packaged semiconductor device, particularly where small pads or pads with a fine pitch are involved. As a result, fiducials are included on the semiconductor device to aid in aligning the placement of the wire bonds and other components with each semiconductor device. During wirebonding, the fiducials are first located, then the position of the bond head may be adjusted or the semiconductor device adjusted (or both) to adjust the placement of the wire bonds and other components to accommodate for variations in the exact position of each individual semiconductor device on the manufacturing line. Fiducials that can be easily and accurately recognized by the wirebonder thus are an important part of the manufacturing process because the precise placement of wire bonds and components with respect to the semiconductor device is important to avoiding failure in the as-assembled semiconductor packages.

[0044] To assist with various alignment algorithms employed by wirebonders, fiducials are often located at the corners or around a perimeter of the semiconductor devices. A one-point fiducial pattern includes using one point in a pattern of the semiconductor die as a reference point, whereas a two-point fiducial pattern includes using two separated points in the pattern as the reference points. In some method implementations, three or more reference points may be employed. While in some circumstances, using a fiducial pattern with one point of reference, such as a gate pad, may be sufficient, fiducial patterns that cover two points of reference physically separated from one another may be more accurate and result in better wire bond and component placement. Two-point fiducial patterns, specifically when located on a diagonal with respect to each other, may provide increased accuracy because the two-point fiducial patterns are able to compensate for die rotation and tilt. Using two-point fiducial patterns in pattern recognition techniques may also further assist with accurate alignment and prevent improper placement of the wire bonds. For some semiconductor devices, some two-point fiducial pattern recognition techniques include using a gate pad as a first fiducial and the “T” intersection of the bus line between two source pads to identify a position and alignment of the semiconductor device including any skew and/or rotation thereof.

[0045] Referring to FIGS. 1 and 2, top views of an implementation of a semiconductor device including a plurality of fiducials are illustrated. Semiconductor device **10** includes a plurality of source pads **12**, **14** adjacent each other and separated by a gap/space, source street **16** that provides electrical isolation between the two pads. Device **10** also includes a gate pad **11** and source runner **18**. Gate pad **11** is disposed adjacent source pads **12**, **14** on one end of semiconductor device **10**. Source runner **18** surrounds source pads **12**, **14** on this side of the semiconductor device and is covered by passivation material, which may be polyimide material in various implementations. In some implementations, the source runner **18** is present at a uniform thickness around a perimeter of source pads **12**, **14**. As illustrated, metal fiducials **13** are located in upper and lower corners at

positions A and C as illustrated by the dotted line regions in FIGS. 1 and 2 with an additional fiducial pattern formed as the entire outline of the gate pad **11** (shown in dotted lines in FIG. 1). Here we have a two-point fiducial pattern between the gate pad fiducial and the metal fiducials **13** on each side of the semiconductor device. As illustrated, in some implementations the metal fiducials **13** are stepped rectangular features at aid in distinguishing their outline from the surrounding metal structure. As previously discussed, semiconductor device **10** may be a power semiconductor device including, by non-limiting example, a metal oxide field effect transistor (MOSFET), an insulated gate bipolar transistor (IGBT), a diode, a thyristor, a silicon controlled rectifier (SCR), or any other kind of power semiconductor device. Source pads **12**, **14**, are used to connect the source of the semiconductor device to the corresponding source interconnect(s) of the semiconductor package. In the particular device illustrated in FIG. 1, the larger the source pads are, the better performance is achieved for the semiconductor package so the size of the source pads is maximized on this side of the semiconductor device. However, this means that little space is reserved for the formation of fiducial structures, particularly metal fiducial structures.

[0046] Referring still to FIGS. 1 and 2, FIGS. 1 and 2 illustrate implementations of fiducials **42**, **44**, **46** arranged on top of source pads **12**, **14**. Fiducials **42**, **44**, **46** are used, either individually or collectively, to align semiconductor device **10** with respect to wire bonding equipment during a wire bonding process to promote desired placement of wire bonds and/or other components. While the use of three different fiducials **42**, **44**, **46** is illustrated in FIGS. 1 and 2, only one of the fiducials, two of the fiducials, or more than three fiducials could be used in other implementations, Fiducials **42**, **44**, **46** may be formed from a photodefinable material that is fully or partially optically transmissive allowing the alignment camera to see through the fiducial while detecting the change in color/pattern the fiducial creates. The photodefinable material may be, by non-limiting example, a photolithographic material, a polyimide, a photosensitive polyimide, or a fully or partially optically transmissive material. In some implementations, the fiducials **42**, **44**, **46** may be a patterned layer over a metal layer and other underlying structures in the semiconductor die. In other implementations, the fiducials **42**, **44**, **46** may be formed of an oxide or nitride material and/or be etched using a pattern formed from a photodefinable material like photoresist.

[0047] In various implementations, the fiducials **42**, **44**, **46** may be arranged over the source pads **12**, **14**, source street **16**, source runner **18** and/or metal fiducials **13**. As illustrated in FIGS. 1 and 2, fiducial **42** is arranged in an upper right corner at position A of semiconductor device **10** and aligned on a diagonal with respect to gate pad **11**. Fiducial **42** overlays source runner **18**, source pad **12** and metal fiducial **13** and has a right triangular geometry and is partially optically transmissive. This results in a change in color of the underlying structures below the structure of the fiducial which the alignment camera of a wirebonder can detect. Fiducial **44** is arranged on an end of source street **16**, opposite gate pad **11** at position B. Fiducial **44** overlays source street **16**, source runner **18** and source pads **12** and **14** and has a substantially equilateral triangular geometry. Fiducial **46** is arranged in a lower right corner at position C of semiconductor device **10** and aligned on a diagonal with respect to gate pad **11**. Fiducial **46** overlays source runner **18**, source pad **14** and metal fiducial **13** and has a right triangular geometry.

[0048] Fiducials **42**, **44**, **46** may be formed of a wide variety of closed shapes that have features that are able to be uniquely identified and can be used in alignment algorithms to allow for calculation of skew and/or rotation of the semiconductor die. For example, while circular shapes could be used, they may not perform as well as straight edged shapes like Y-shaped features, crosses, concave polygons, star-shaped features and other. However, where the shape of the fiducial is used in combination with another adjoining or underlying structure that helps provide an orientation like the source running, the use of convex polygons and elliptical shapes could also be used. In various implementations, the same or different fiducial shapes may be employed where multiple fiducials are used.

[0049] When using a multi-point pattern recognition method or technique that includes gate pad **11**

as one reference pattern/point, positioning at least one additional fiducial **42, 44, 46** (or the group of two or all three of the fiducials) as the second reference pattern/point across from gate pad **11** instead of, for example, on a same side as gate pad **11**, may provide increased accuracy and better bonding results due to increased ability to detect rotation/skew of the semiconductor device. In addition, when using a multi-point pattern recognition method or technique that includes gate pad **11** as one point/pattern, positioning a second fiducial on a diagonal with respect to the first point/pattern, gate pad **11**, may also provide increased alignment accuracy and thus better bonding results. Thus, in some implementations, using fiducials **42, 46** at positions A or C in conjunction with gate pad **11** in a two-point pattern may be beneficial. Further, when using a three-point pattern recognition method or technique that includes three fiducials, arranging the fiducials in a triangular shape may also provide increased accuracy and better bonding results. Thus, in further implementations, using fiducials **42, 46** at positions A and C in conjunction with gate pad **11** in a three-point triangular pattern may be beneficial.

[0050] In some implementations, fiducials **42, 44, 46** are created using photolithography and masking methods from photodefinable materials. A layer of photodefinable material, photosensitive material, photoresist, or other material may be deposited on the semiconductor device **10** during the end of the semiconductor fabrication process. A mask, including the pattern for the desired fiducials is used to expose the photodefinable material and light is transmitted through the mask. The pattern of the mask is transferred into the photodefinable material which is then developed to remove the exposed or unexposed portions (depending on the type of photodefinable material used) leaving behind the desired fiducials. Additional baking and/or curing steps may be employed to ensure the photodefinable material is fully stabilized and ready to remain on the semiconductor die through the packaging process. In other implementations, oxides or nitrides that are not be photodefinable themselves are initially deposited on semiconductor device **10**, and subsequently patterned using photodefinable materials like photoresist followed by etching processes resulting in fiducials with a pattern etched into the material of the oxide/nitride.

[0051] As illustrated in FIGS. **3** and **4**, a portion of a semiconductor device **110** comprising a source pad **114**, source runner **118**, a source or metal layer **120** is illustrated. As illustrated in FIG. **4**, the semiconductor device has a semiconductor substrate that may have doped regions formed within the substrate to form an active (p) layer **130** and a passive (n) layer **132**. Gates **122** are arranged between channels **126** and sources/drains **124**. When a positive voltage is applied to gate **122**, negative electrons flow from the source **124** and drain through channels **126** allowing a current to pass through. Electricity flows from the source to the drain during operation of the device. As illustrated in FIG. A metal fiducial **113** is provided in a lower corner of the semiconductor device **110**. The metal fiducial **113** is shown having a stepped geometry formed by stepping the corner of the source pad, creating a corresponding space that is darker when viewed by the camera than the more reflective metal.

[0052] One solution for improving wire bond setting include enlarging metal fiducial **113** on the die corner for better wire bond pattern recognition by making the space adjoining the source pad larger. However, enlarging the metal fiducial reduces the size of the source metal layer **120** which can affect electrical performance because the electrical structure (gates/channels/drains) underneath the source metal layer **120** may be exposed which can create reliability risk. In addition, changing the size of the metal fiducial includes changing the mask used to create the metal fiducial **113** which is itself a costly process.

[0053] In contrast, FIGS. **3** and **4**, illustrate forming a fiducial **140** using a photodefinable material deposited over source pad **114**, specifically, over source or metal layer **120**. As illustrated in FIG. **3**, because the photodefinable material is partially optically transmissive, it creates a different colored/contrasting pattern at the corner of the source pad that complements or works with the existing fiducial shape. In some implementations, the use of the fiducial **140** may replace or eliminate the need to use the metal fiducial **113**.

[0054] In some implementations, the photodefinable fiducial is a photosensitive material such as a polyimide, for example, which is already patterned on the surface of the semiconductor device where the fiducial **140** is formed by adjusting the mask pattern used to form the polyimide layer. As illustrated in FIGS. **3** and **4**, metal fiducial **113** provides a fiducial width of $W_{sub.113}$ available for pattern recognition detection. By applying photodefinable fiducial **140** over metal layer **120**, an additional width, width $W_{sub.140}$, is available for pattern recognition detection in addition to width $W_{sub.113}$. Thus, a total width available for pattern recognition is increased from width $W_{sub.113}$ by with $W_{sub.140}$ to a total fiducial width, width $W_{sub.T}$. The increase in the width comes with an increase the total area of the fiducial and the additional of an additional shape(s) that the alignment camera can use during pattern recognition. Also, applying a photosensitive fiducial **140** over top of the source pad layer **120** increases the width/area/shape available for pattern recognition without altering or changing the underlying active area, source pad layer **120**, source pad **114**, or electrical parameters of semiconductor device **110**. The increased width/area/shape improves detection and analysis by pattern recognition techniques and modalities. The photodefinable layer including fiducial **140** may be formed on the surface of semiconductor **110** at the last or one of the last process steps, thus making fiducial **140** easy and simple to fabricate without affecting the function and reliability of semiconductor device **110**. In addition, fiducial **140** may be deposited over an active region of the source pad as illustrated in FIG. **4** without impacting the electrical characteristics of the semiconductor device **110**.

[0055] FIGS. **5** to **7** illustrate portions of semiconductor devices **210** with fiducials **240**, **241** formed on metal layers **220**, **221** of source pads **212**, **214** respectively. Source pads are separated by a space and/or a non-active area, source street **216** which acts to electrically insulate the source pads **212**, **214** from each other. Source runner **218** surrounds a perimeter of source pads **212**, **214**. Each source pad **212**, **214** includes metal layer **220**, **221**, respectively. Fiducials **240**, **241** are made of a photodefinable material which, in some implementations, may include photosensitive materials such as a polyimide, or any other photodefinable material type disclosed in this document. As illustrated in FIGS. **5** to **7**, fiducials **240** are located in a lower corner of source pads **212** and fiducials **241** are located in an upper corner of source pads **214**. Fiducials **240**, **241** are illustrated as being rectangular or square in shape but can include any desired geometry. Fiducial **240** includes a length $L_{sub.240}$ and a width $W_{sub.240}$, source street includes a length $L_{sub.216}$ and source runner **218** includes a width $W_{sub.218}$. As illustrated in FIG. **5**, fiducial **240** includes a width $W_{sub.240}$ of 200 microns and a length $L_{sub.240}$ of 200 microns, source runner **218** has a width.sub.218 of 73 microns and source street **216** has a length of $L_{sub.216}$ of 82 microns. In some implementations, a single fiducial pattern used for alignment may encompass fiducials **240**, **241** and/or a portion of source street **216** having the photodefinable material formed thereon. In various implementations, a single fiducial pattern used during alignment may encompass fiducials **240**, **241** and the areas of source runner **218** and/or source street **216** having the photodefinable material deposited thereon resulting in a fiducial having a width that is 275 micrometers wide or less and a length of that is 500 microns or less. In some implementations, the fiducials **240**, **241** overlap the source pad by 200 microns in width and/or length. In some implementations, fiducials **240**, **241** may be utilized as one fiducial extending across two source pads **212**, **214** and source street **216**. The dimensions of the fiducial structures here are merely for the exemplary purposes of this disclosure as smaller or larger fiducials are possible using the principles disclosed herein.

[0056] FIGS. **6** and **7** illustrate fiducials **240**, **241** that include metal alignment features **242**, **243** as part of the fiducial structure. The metal alignment features **242**, **243** may be located within a perimeter or of fiducials **240**, **241**, may be spaced apart from a perimeter of fiducials **240**, **241** as illustrated in FIG. **6**, or aligned with at least one edge of fiducials **240**, **241** as illustrated in FIG. **7**. FIG. **6** illustrates metal alignment features **242**, **243** floating within a perimeter of fiducials **240**, **241** respectively, such that the metal alignment features **242**, **243** are spaced apart from each edge of fiducials **240**, **241**, respectively. In some implementations, the metal alignment features **242**, **243**

may be floating within fiducials **240**, **241**, respectively, and, for example, may be spaced apart from each edge of fiducials **240**, **241** by 50 microns or less though larger or smaller distances could be used in various implementations.

[0057] FIG. 7 illustrates metal alignment features **242**, **243** in which the metal alignment features **242**, **243** border two edges of fiducials **240**, **241**, respectively and photodefinable material borders two edges of fiducials **240**, **241**, forming part of the same layer of photodefinable material that forms the fiducials **240**, **241**. In some implementations, metal alignment features **242**, **243** may be spaced apart from at least one edge of fiducials **240**, **241** by 50 microns or less. In various implementations, a metal alignment feature **242**, **243** may border at least one edge of a fiducial **240**, **241**, respectively.

[0058] A length and a width of metal alignment features **242**, **243** maybe 150 microns or less in a particular implementation. The metal alignment features **242**, **243** may be any shape disclosed herein for a fiducial. Here the shape of the metal alignment features is square. In some implementations, metal alignment features **242**, **243** may be made of the source metal layer **220**, **221**, respectively whereas in other implementations, metal alignment features **242**, **243** may be formed of a separate layer of metal. As illustrated in FIGS. 5-7, the metal alignment features **242**, **243** are optically visible through the material of their respective fiducials. Photolithography and masking methods may be used to form fiducials **240**, **241** having the desired metal alignment features **242**, **243** either at the same time as the source metal layer or afterward.

[0059] FIG. 8 illustrates a semiconductor device **310** comprising fiducials **340**, **341** formed on metal layers **320**, **321** of source pads **312**, **314** respectively. Source pads **312**, **314** are separated by a space and/or a non-active area, source street **316** used to provide electrical isolation between the source pads **32**, **314**. As illustrated, source runner **318** surrounds a perimeter of source pads **312**, **314**. Each source pad **312**, **314** includes/is composed of a metal layer **320**, **321**, respectively. Fiducials **340**, **341** are comprised of a photodefinable material which, in some implementations, includes photosensitive materials such as polyimide or any other photosensitive material like those disclosed in this document. As illustrated in FIG. 8, fiducial **340** is located in a lower corner of source pad **312** and fiducial **341** is located in an upper corner of source pad **314**. Fiducials **340**, **341** are illustrated as being triangular in shape but can include any shape disclosed in this document. Here they take the form of a right triangles. Fiducial **340** includes a length $L_{sub.340}$ and a width $W_{sub.340}$, the length $L_{sub.340}$ and width $W_{sub.340}$ may each be 200 microns or less in a particular implementation. What is different about the fiducials **340**, **341** of FIG. 8 from those in FIGS. 6 and 7 is that there is no patterning of the metal layer of the source pads **312**, **314**, but the shape of the fiducial is formed just using the photodefinable material or patterned oxide/nitride layer. This is a similar approach to the fiducials **240**, **241** of FIG. 5, which are rectangular.

[0060] FIGS. 9 and 10 illustrate top views at different magnifications of the implementation of fiducial **440** formed over source pads **412**, **414** of a semiconductor device **410**. Semiconductor device **410** includes gate pad **411**, source pads **412**, **414** separated by gate runner **416**. Source runner **418** also surrounds a perimeter of source pads **412**, **414**. As illustrated in FIG. 9, fiducial **440** is formed of a pattern formed in a photodefinable material deposited over source pads **412**, **414**, gate runner **416** and source runner **418** that equipment may be like any disclosed in this document, but in this case is in the form of two right triangles opposing each other across the width of the gate runner **416**. Fiducial **440** includes a triangular shape overlapping source pads **412**, **414** which, along with the shape of a portion of the source street contrasting with the source pad material, results in better wire bond placement than a “T” shaped fiducials formed at the intersection of the source street and source runner because this shape is more reliably recognized by the image recognition software than the shape of just the corners of the source pads and portion of the source street alone. Fiducial **440** may be formed using photolithography and masking techniques disclosed herein for use in forming photodefinable materials and oxides/nitrides.

[0061] In places where the description above refers to particular implementations of semiconductor

devices and implementing components, sub-components, methods and sub-methods, it should be readily apparent that a number of modifications may be made without departing from the spirit thereof and that these implementations, implementing components, sub-components, methods and sub-methods may be applied to other semiconductor devices.

Claims

1. A semiconductor device comprising: a first die pad having a first metal layer thereon; and a fiducial for aligning to the semiconductor device, the fiducial comprising a portion comprising a photodefinable material formed over the first metal layer of the first die pad.
2. The semiconductor device of claim 1, further comprising: a second die pad adjacent the first die pad, the second die pad having a second metal layer deposited thereon; and a second fiducial formed over the second metal layer.
3. The semiconductor device of claim 2, further comprising: a space between the first die pad and the second die pad wherein the second fiducial includes the space.
4. The semiconductor device of claim 1, wherein the fiducial is formed over an active portion of the first die pad.
5. The semiconductor device of claim 1, further comprising: a second fiducial for aligning to the semiconductor device, the second fiducial comprising a photodefinable material formed over the first metal layer of the first die pad.
6. The semiconductor device of claim 1, wherein the fiducial is triangular.
7. The semiconductor device of claim 1, wherein the fiducial comprises a metal alignment feature.
8. The semiconductor device of claim 7, wherein the metal alignment feature is part of the first metal layer.
9. The semiconductor device of claim 7, wherein the metal alignment feature is square.
10. The semiconductor device of claim 7, wherein the metal alignment feature is stepped.
11. The semiconductor device of claim 7, wherein the metal alignment feature is rectangular.
12. The semiconductor device of claim 7, wherein the metal alignment feature is triangular.
13. The semiconductor device of claim 7, wherein the metal alignment feature is spaced apart from an edge of the fiducial.
14. The semiconductor device of claim 7, wherein the photodefinable material borders at least one edge of the fiducial.
15. The semiconductor device of claim 14, wherein the photodefinable material borders at least two edges of the fiducial.
16. A semiconductor device comprising: a first die pad comprising a first metal layer; and a second die pad adjacent the first die pad, the second die pad comprising a second metal layer; and a first fiducial for aligning the semiconductor device, the first fiducial including a photodefinable material formed over one of the first metal layer, the second metal layer, or both the first metal layer and the second metal layer.
17. The semiconductor device of claim 16, wherein the first fiducial includes a portion of the first metal layer and a portion of the second metal layer.
18. The semiconductor device of claim 16, further comprising: a second fiducial for aligning the semiconductor device, the second fiducial including a photodefinable material formed over the first metal layer, the second metal layer, or both the first metal layer and the second metal layer.
19. The semiconductor device of claim 18, further comprising: A third fiducial for aligning the semiconductor device, the third fiducial including a photodefinable material formed over the first metal layer, the second metal layer, or both the first metal layer and the second metal layer.
20. A semiconductor device comprising: a first die pad; and a second die pad adjacent the first die pad; and a first fiducial for aligning the semiconductor device, the first fiducial including a material

formed over one of the first die pad, the second die pad, or both the first die pad and the second die pad.
